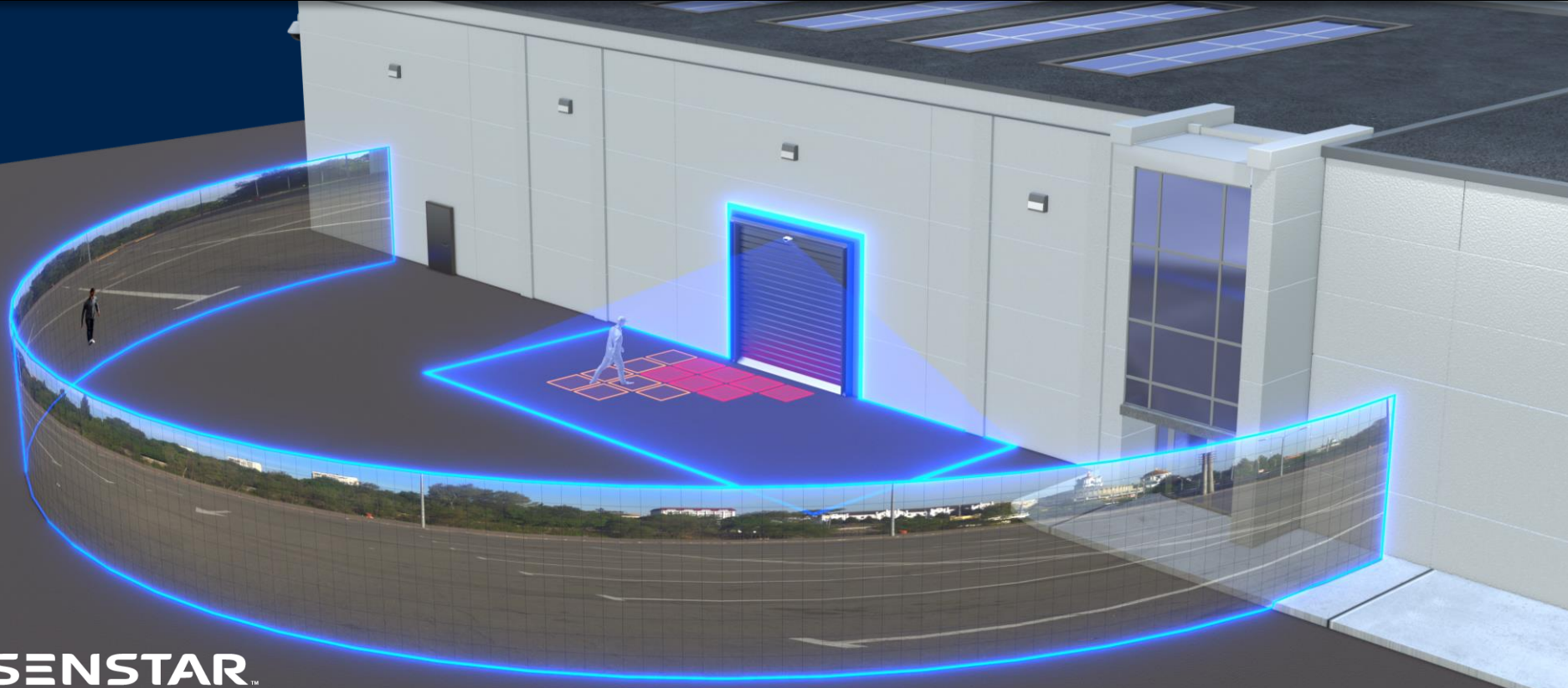


Senstar MultiSensor™

AI-POWERED MULTI-SENSING DETECTION SYSTEM



SENSTAR™

The Security Market Challenge

Nuisance Alarm Rates

Customers prioritize security devices with lower nuisance alarm rates, emphasizing reliability through minimized false alerts.

VS.

Situational Awareness

Growing demand for security devices to provide situational awareness giving more context from alarms.

The Solution: Senstar MultiSensor™

Comprehensive Threat Detection

AI-Based Sensor Fusion Engine defeats nuisance alarms

Enhanced Situational Awareness

Configurable pre-alarm/alarm zones provide context for each alarm

Rugged, Innovative Design

Reduce labor costs with easy installation and maintenance

Flexible, Scalable Solution

Integrate with existing security infrastructure, scale as required

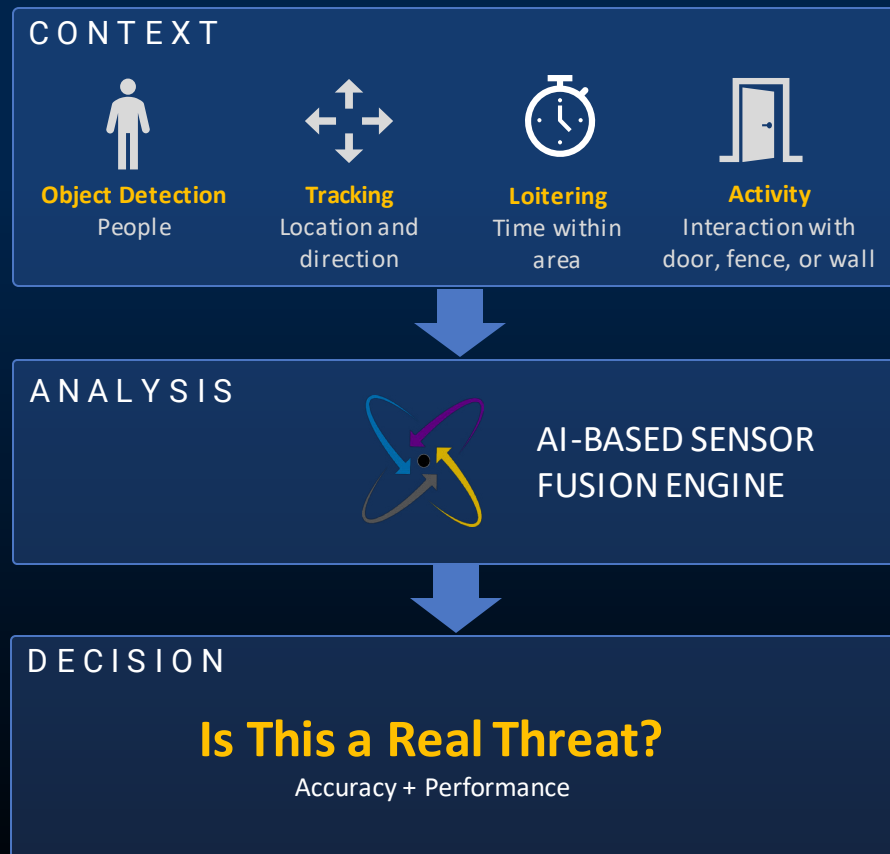
Multiple Integration Options

Integrate with popular video and security management systems



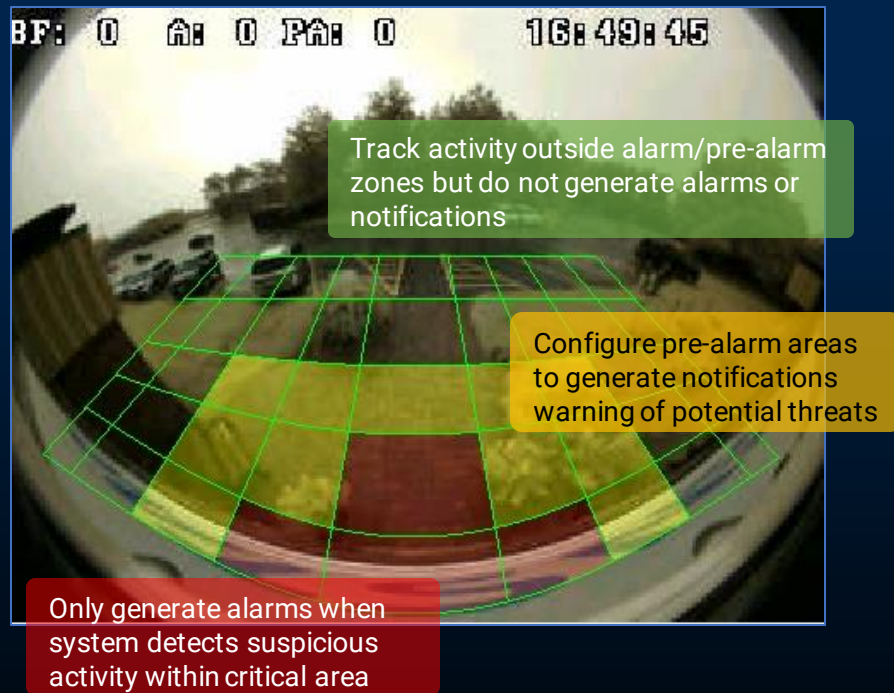
Comprehensive Threat Detection

- Data from multiple sensors is synthesized in real-time by AI-based Sensor Fusion Engine
- Like the human brain and our different senses, the Sensor Fusion Engine uses the different types of data to understand the context in which the event is occurring
- Sensor fusion achieves benefits of different sensing technologies while avoiding their limitations, specially reject disturbances caused by:
 - Rain, snow and fog
 - Vehicle headlights and shadows
 - High winds or location-specific vibrations



Enhanced Situational Awareness

- Separate protected area into pre-alarm and alarm areas
- Avoid distracting nuisance alarms caused by activity outside critical area
- Provide meaningful, correct information to security personnel:
 - Boots-on-the-ground view of threat
 - Alarm/pre-alarm location
 - Link alarms to high-resolution cameras for further investigation



Rugged, Innovative Design

- Robust design and weather-resistant housing
- Maintenance-free (enclosure design includes sun visor and rain channels)
- Integrated mounting bracket with security screws for quick and easy installation
- Suitable for **outdoor** and **indoor** applications

Specs at a Glance

Temperature: -40 to +60 °C

Ingress protection: IP54

Power (PoE): 3W (max)

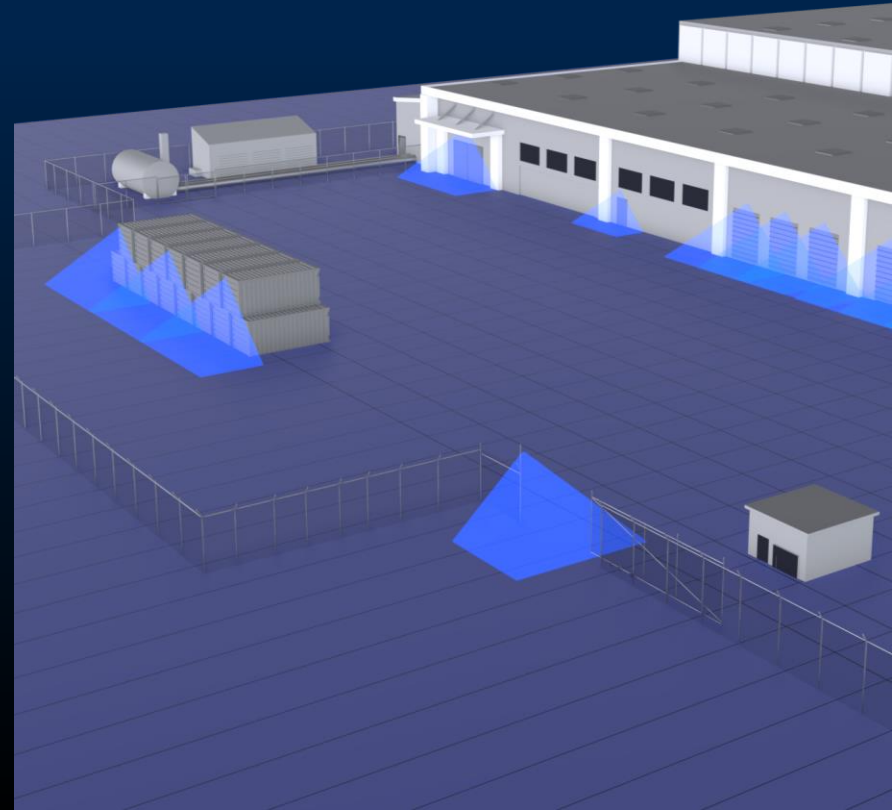
Weight: 0.6 kg (1.3 lbs)

Size: 18 x 19 x 10 cm (7 x 8 x 4 in)



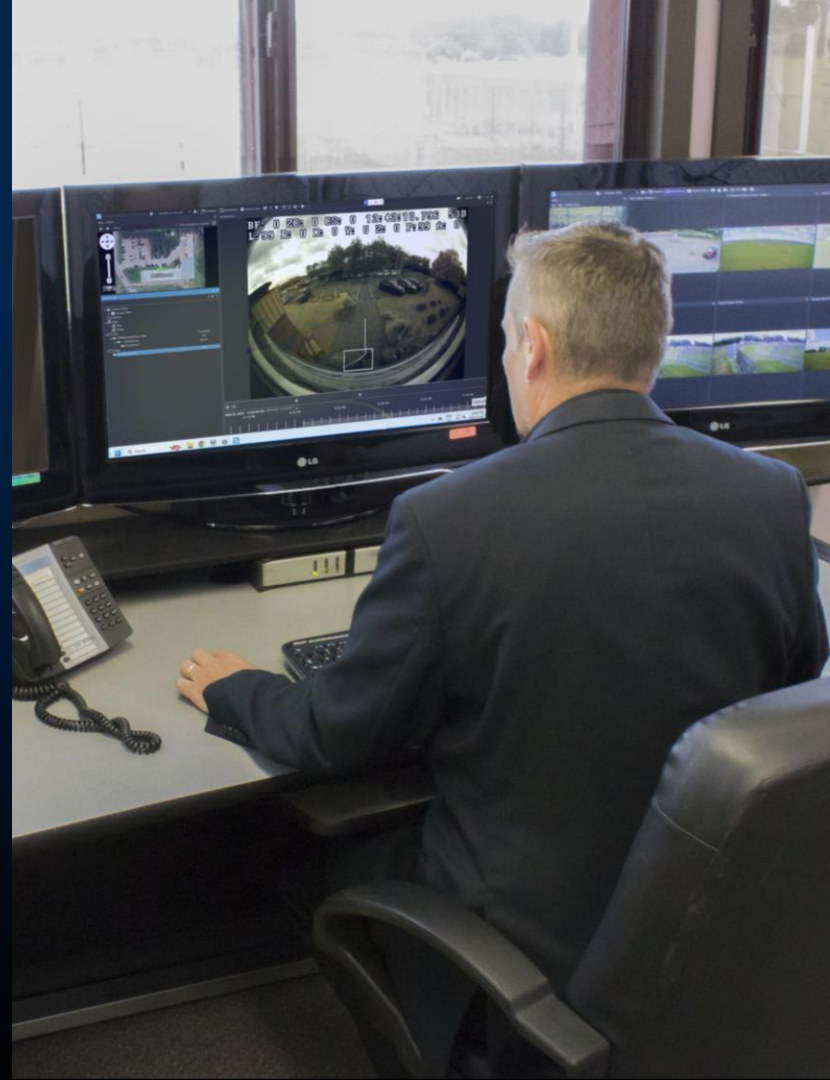
Flexible, Scalable Solution

- Scalable PoE design enables deployment of any number with minimal additional infrastructure (re-use existing camera network)
- Use in a wide range of applications:
 - Building entrances
 - Perimeters
 - Gates and sally ports
 - Tower structures
 - Open areas
- Deploy anywhere that requires intrusion detection



Multiple Integration Options

- Easily add to existing security infrastructure
- Interface with virtually any security system
- Software integrations available for popular VMS/SMS applications
- View video via any RTSP client



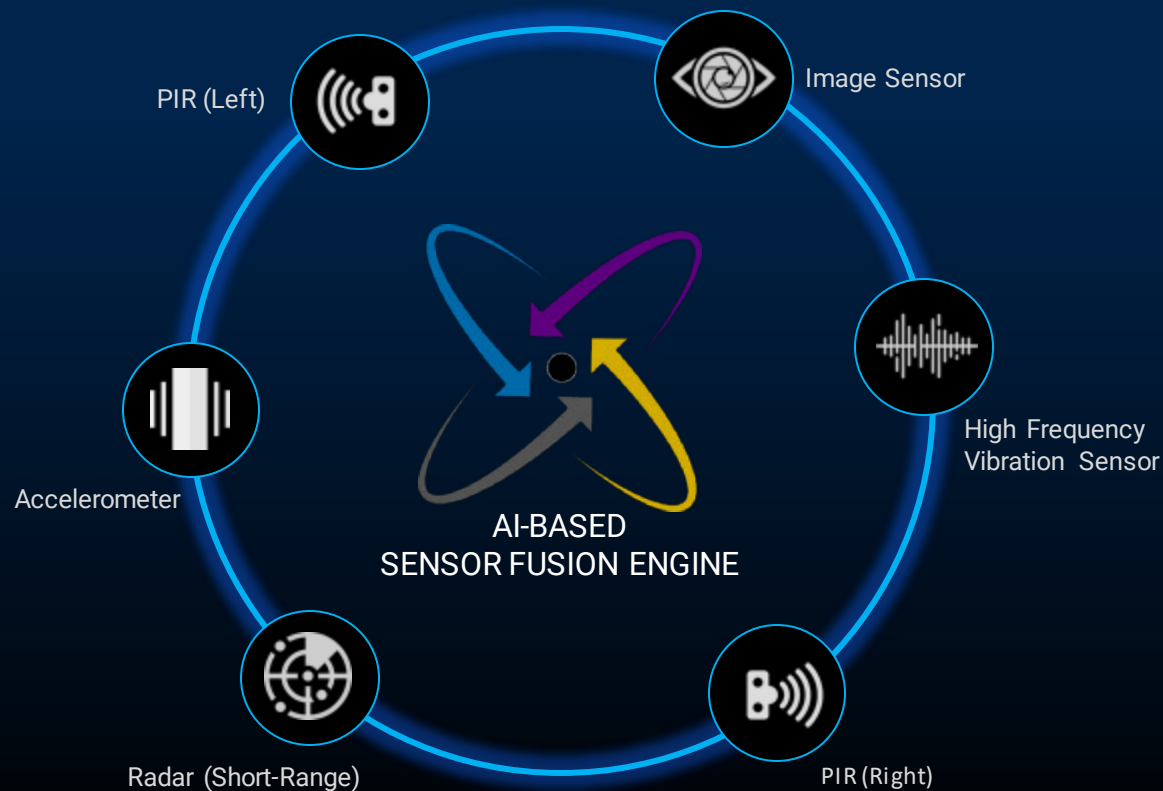


Technology

The Technology



Senstar Sensor Fusion Engine



Process raw sensor
input in **real-time**

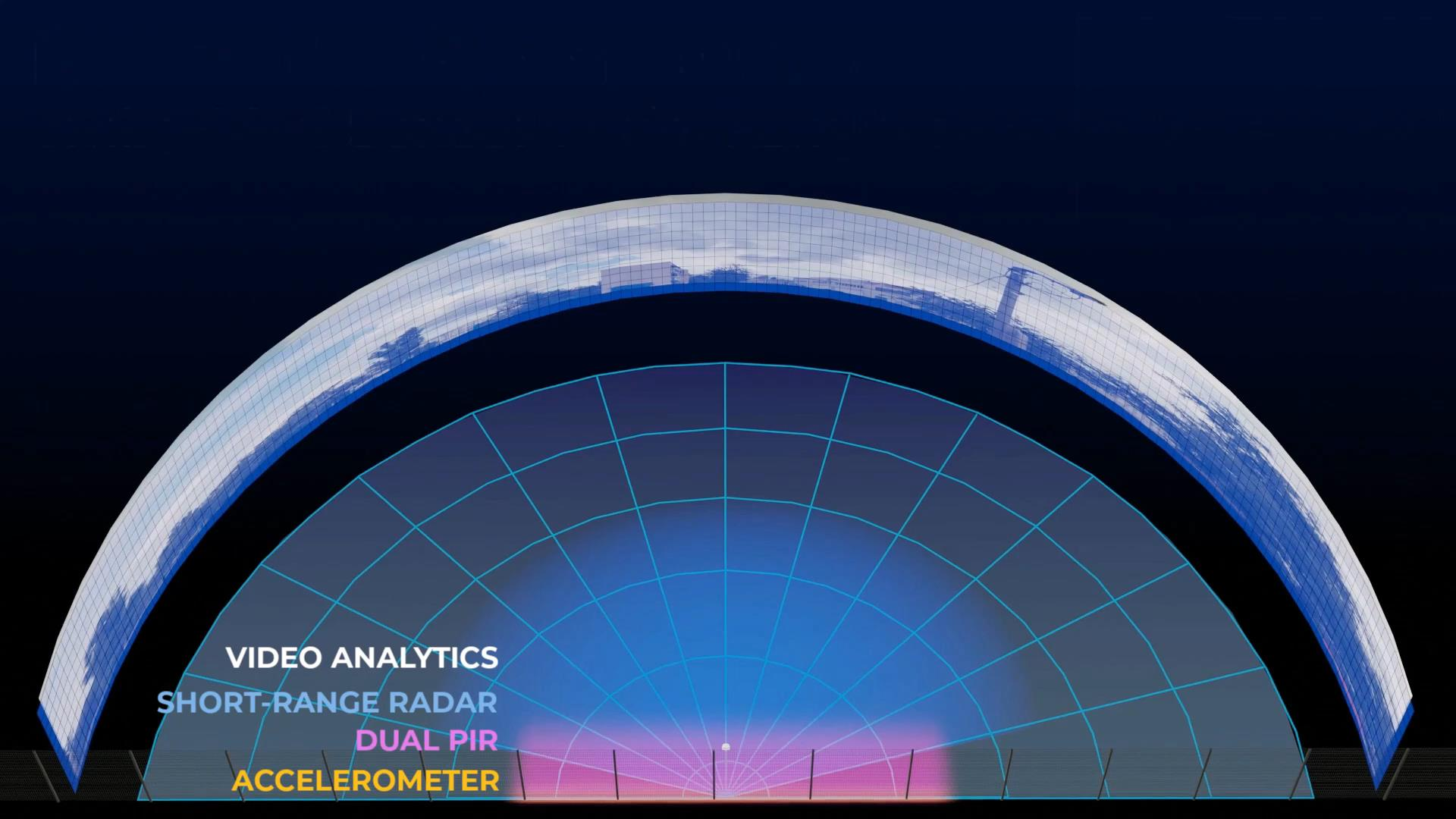
with

AI-based
intelligence

for

**Unmatched
performance
and accuracy**





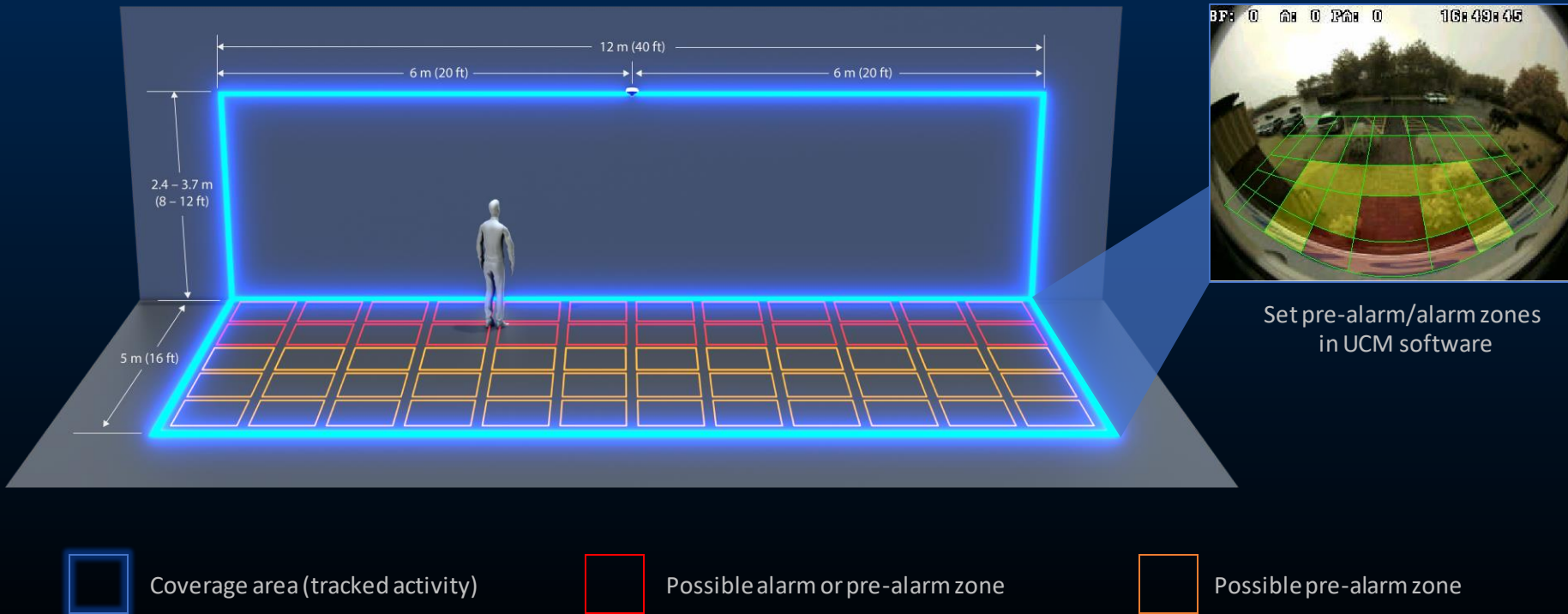
VIDEO ANALYTICS
SHORT-RANGE RADAR
DUAL PIR
ACCELEROMETER



Performance

Coverage Area

Localized coverage along protected area enables system to focus on real threats, not nearby activity. Alarm and pre-alarm zones are configurable in software.



Detection Performance: Intrusion Attempt (Day)

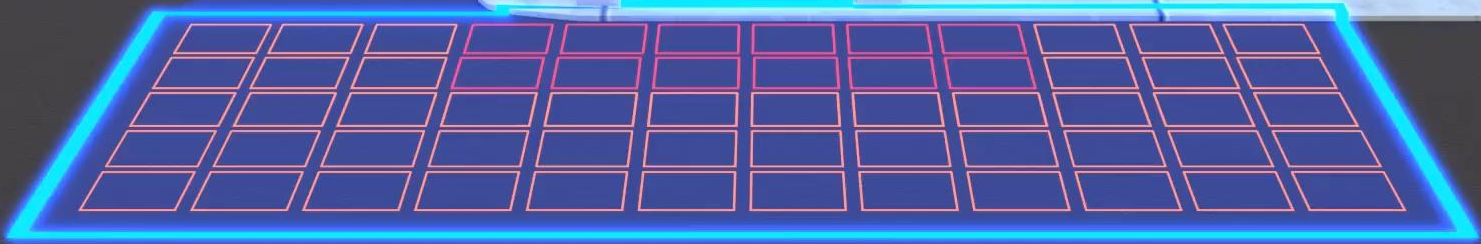


Intrusion attempts happen at all hours of the day:

- Person is detected, accurately tracked, and alarm notification sent
- As person attempts to open the building door an alarm is triggered, and notification sent

Second row camera decorations were included for test purposes; deployed systems can hide advanced decorations.

BF: 50° RC: 0 CS: 0 12:55:15.132 550
L: 0 R: 0 X: 0 Y: 0 Z: 0 P: 0 A: 0

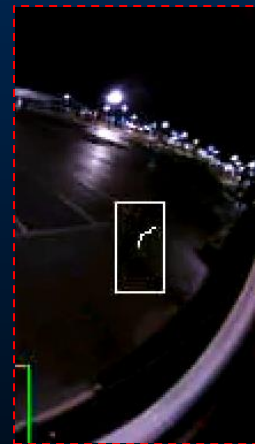


Detection Performance: Potential Stealth Attack (Night)



Suspicious activity at night in an area with poor lighting:

- Person is accurately tracked despite poor lighting that reduces the effectiveness of video analytics
- As person does not attempt intrusion, there is only pre-alarm tracking, not an actual intrusion alarm



Second row camera decorations were included for test purposes; deployed systems can hide advanced decorations.

Detection Performance: Stealth Attack



Intrusion attempt by sophisticated intruder:

- Comprehensive camouflage (attempt to fool video analytics)
- Slow movement (attempt to trick sensors)

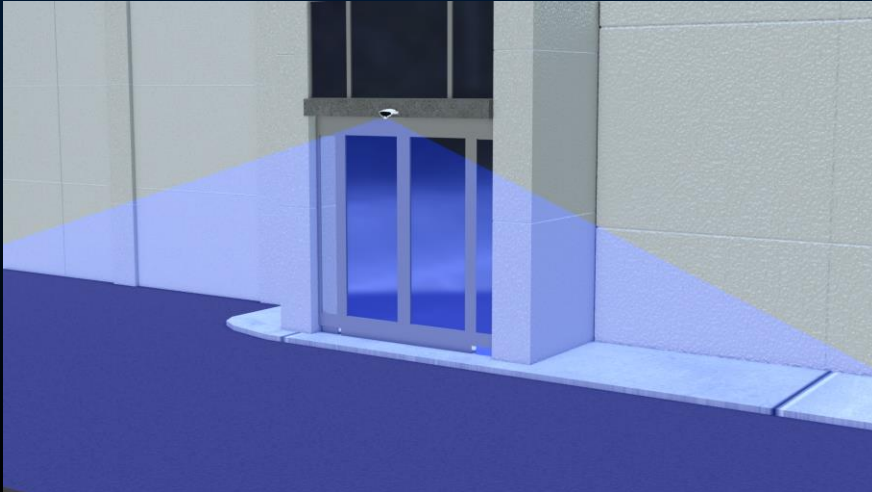
Second row camera decorations were included for test purposes; deployed systems can hide advanced decorations.



Applications

Entrances

- Protect swing and sliding doors, garage doors, and glass store fronts
- Avoids nuisance alarms generated by:
 - People walking past entrance
 - Cars parked nearby
 - People exiting the building



Fences and Gates

- Protect fences of all types, including chain link, welded mesh, expanded metal, and palisade
- Detects any attempt to cut, climb, lift, or otherwise bypass the fence fabric
- Avoids nuisance alarms generated by:
 - High winds, heavy rain, and blowing debris
 - People walking past area
 - Cars parked nearby
 - Gate activity (opening, closing)
 - People exiting gates



Walls

- Protect walls of all construction types (concrete, metal, brick, stone, etc) from climbing, cutting, or smashing
- Avoids nuisance alarms generated by:
 - People walking nearby
 - Cars parked nearby



Open Areas

- Senstar MultiSensor provides intelligent detection in areas without a vibration-generating barrier
- Includes staircases, hallways, open entrances, and ladders
- Avoids nuisance alarms generated by:
 - People walking nearby
 - Cars parked nearby

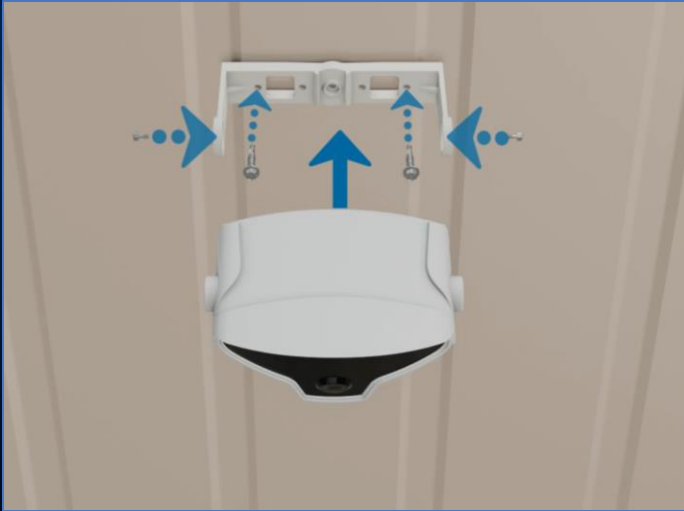




Installation and Configuration

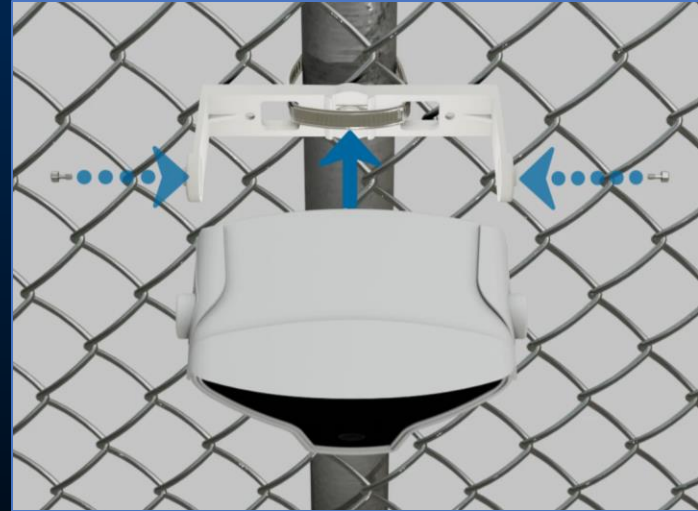
Installation Options

Wall Installation



- Use surface-appropriate mounting hardware
- Select **Barrier-Free** option in UCM

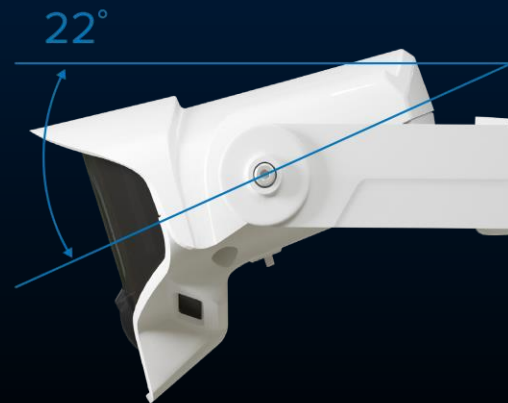
Post Installation



- Fence or free-standing
- Use stainless steel gear clamp
- Disable **Barrier-Free** option in UCM

Height and Tilt Requirements

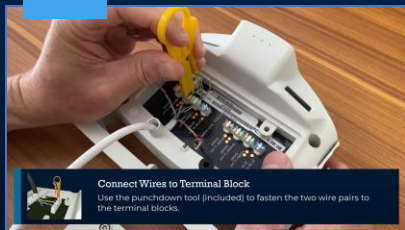
- Install at a height between 8 and 12 ft (2.4 to 3.7 m)
- Tilt downwards at 22° (use built-in guide)
- Set height and review tilt in UCM software



Installation and Configuration

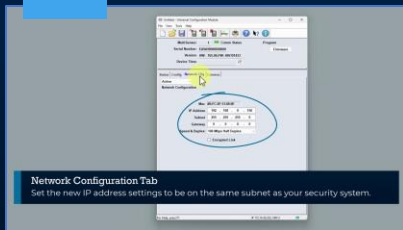
Installation is similar to that of an IP camera:

1



Connect device to
PoE switch

2



Configure IP address
in UCM

3



Install on wall, fence
or post

4



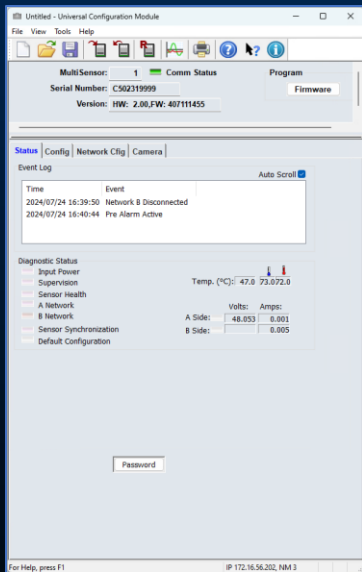
Configure detection
parameters

For More Info

How-to videos and help [available online](#)

Configuration and Operation

Configuration parameters are set in UCM software:



Status information
Event history



Yellow: Pre-alarm area
Red: Alarm area



Network settings



Camera settings
Device LED visibility



Networking and Integration

Deployment Configurations

Senstar MultiSensor™ deployment configurations are both simple and flexible:



SENSTAR®

Symphony Common
Operating Platform

Genetec™

Security Center

 **milestone**

XProtect (all versions)



ASCII strings
via NMS



Alarms only

Deployment Options

Senstar MultiSensor™ configuration is both simple and flexible:

- Senstar Symphony Common Operating Platform
- Genetec / Milestone VMS (NMS Gateway)
- Alarm Management via ASCII string from NMS
 - SMS/PSIM
 - VMS that supports ASCII string input (Avigilon, Hanwha, Bosch, Exacq, etc. etc.)
- Relay output (alarms only)

Deployment Options

Required Components



Senstar MultiSensor™

Ethernet cable connected to PoE Switch



PoE Switch

Power and networking over a single cable



Universal Configuration Module (UCM)

Used to configure & update firmware
Free download from www.senstar.com



Network Manager Service (NMS)

Gateway for MultiSensor to alarm display interface
Free download from www.senstar.com

Optional Items



Senstar Symphony License

Display alarms and video in Senstar Symphony



Gateway Integration

Receive alarms into Genetec, Milestone, Lenel systems



ASCII alarm strings from NMS

Receive alarm and status information by any application capable of receiving ASCII strings



MultiSensor relay output

Trigger a device using the built-in relay output



RTSP streams from NMS

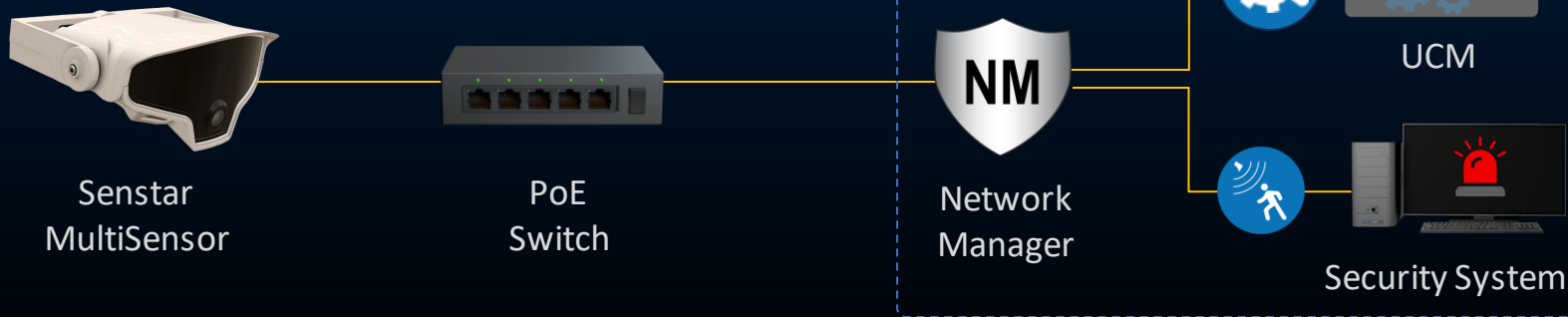
View video with any video system supporting RTSP

Networking Overview

Senstar MultiSensor uses Ethernet for communications, with the Network Manager software (installed on PC or server) acting as a gateway:

- **Alarms:** Forwarded to site's security system (multiple options)
- **Video:** Processed and streamed (RTSP) to video system
- **Configuration:** UCM connects to device via Network Manager

Network Manager software is available in a **free MultiSensor-only version** as well as the standard Senstar multi-device license version



Alarm Outputs

Senstar MultiSensor outputs alarms in a variety of formats:

- **Relay output:** Each device includes an alarm relay output
- **ASCII string:** Custom strings transmitted over TCP/IP
- **Integrations (tested):**
 - Senstar Symphony Common Operating Platform
 - Genetec Security Center (via NMS-GSC Gateway)
 - Milestone XProtect (via NMS-XProtect Gateway)

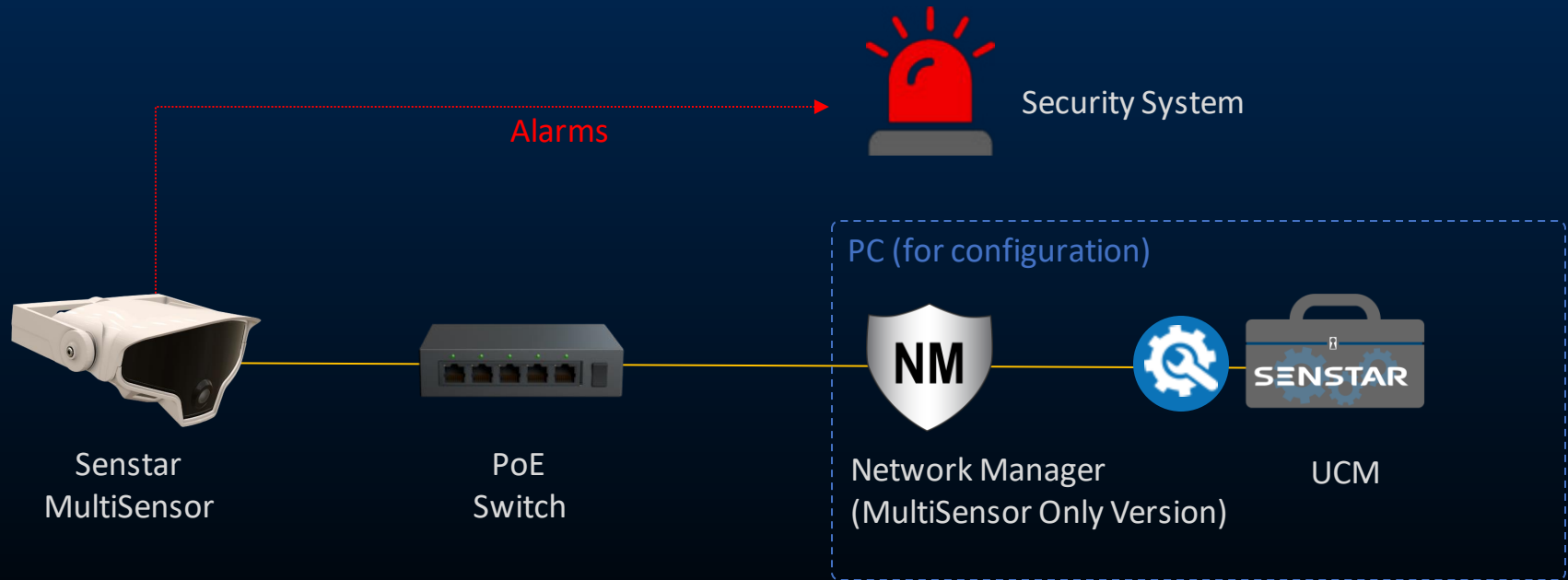
Have Another System?

You can connect to virtually any security system via
Senstar's hardware and software ecosystem
(contact Senstar for details)



Relay Output Only

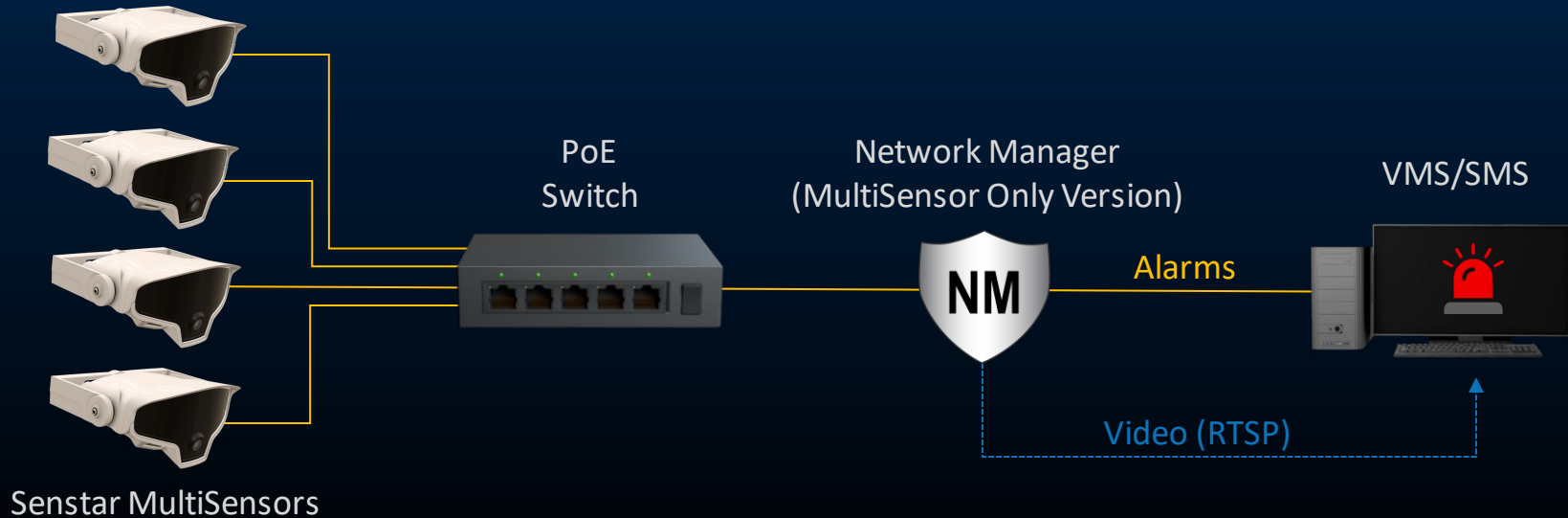
Senstar MultiSensor includes a solid-state relay output (100mA@60V) for transmitting alarms.



Integration –MultiSensor-Only Systems

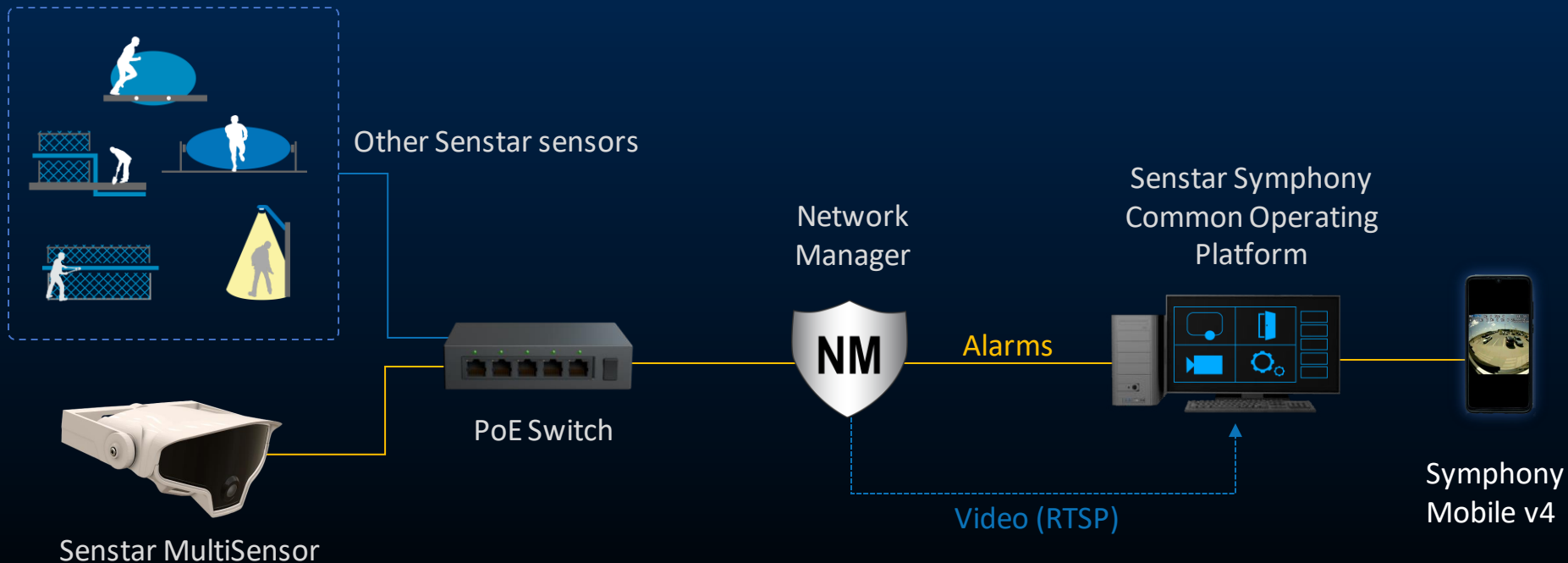
Multiple Senstar MultiSensor devices may be connected in a network (star topology):

- The free MultiSensor-only version of Network Manager may be used
- If the devices are installed close together, different radar channels should be configured in UCM software



Integration – Senstar Symphony

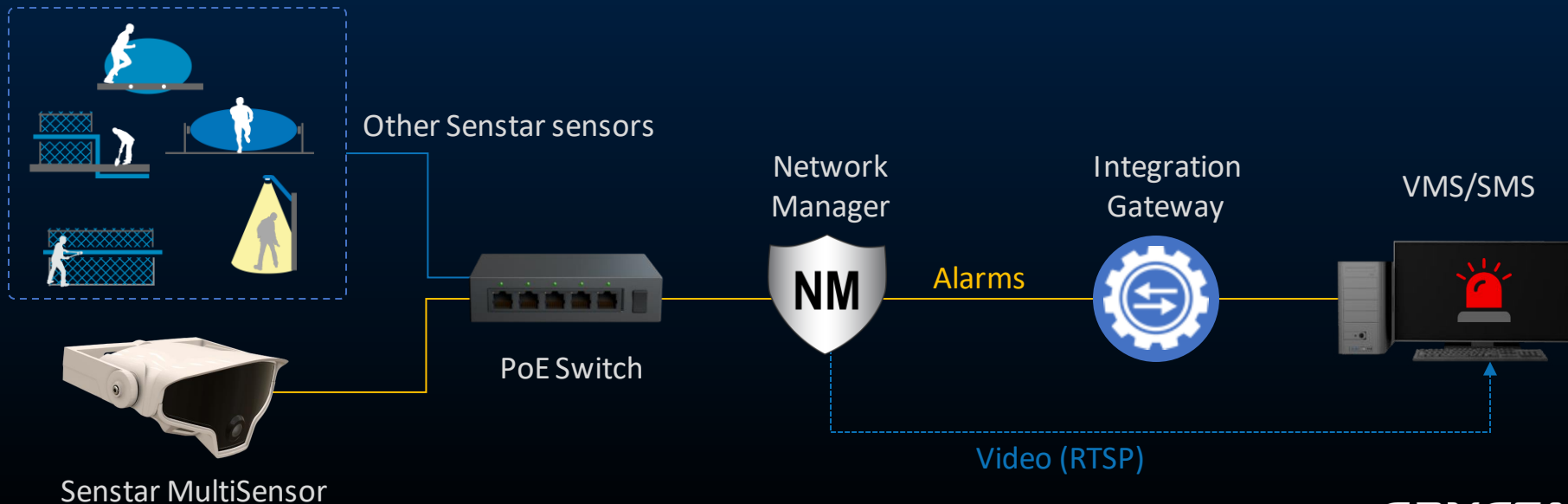
The Senstar Symphony integration lets the user view alarms and video via a single license.



Integration – Genetec or Milestone

Gateway integrations with Genetec and Milestone are tested and available. Note:

- Network Manager Gateway only passes alarm information, not video.
- User running Genetec, Milestone or other VMS may require camera license to display RTSP video
- If system only has Senstar MultiSensor devices (no other Senstar sensors), the free version of the Network Manager software may be used



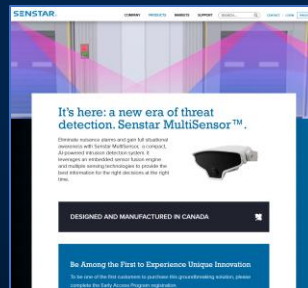


Summary and Discussion

For More Information



Datasheet



Website



Quick Start Guide



How-To Videos



Marketing Video